

Figure 9.15: An elliptical path.

follows the presentation in [15] because of its conceptual simplicity.

Other research has focused on numerical methods such as dynamic programming or nonlinear optimization to minimize cost functions such as actuator energy. One early example of work in this area is by Vukobratović and Kirčanski [191].

9.8 Exercises

Exercise 9.1 Consider an elliptical path in the (x, y) -plane. The path starts at $(0, 0)$ and proceeds clockwise to $(2, 1)$, $(4, 0)$, $(2, -1)$, and back to $(0, 0)$ (Figure 9.15). Write the path as a function of $s \in [0, 1]$.

Exercise 9.2 A cylindrical path in $X = (x, y, z)$ is given by $x = \cos 2\pi s$, $y = \sin 2\pi s$, $z = 2s$, $s \in [0, 1]$, and its time scaling is $s(t) = \frac{1}{4}t + \frac{1}{8}t^2$, $t \in [0, 2]$. Write down $\dot{X}$ and $\ddot{X}$.

Exercise 9.3 Consider a path from $X(0) = X_{\text{start}} \in SE(3)$ to $X(1) = X_{\text{end}} \in SE(3)$ consisting of motion along a constant screw axis. The path is time scaled by some $s(t)$. Write down the twist $\mathcal{V}$ and acceleration $\dot{\mathcal{V}}$ at any point on the path given $\dot{s}$ and $\ddot{s}$.

Exercise 9.4 Consider a straight-line path $\theta(s) = \theta_{\text{start}} + s(\theta_{\text{end}} - \theta_{\text{start}})$, $s \in [0, 1]$ from $\theta_{\text{start}} = (0, 0)$ to $\theta_{\text{end}} = (\pi, \pi/3)$. The motion starts and ends at rest. The feasible joint velocities are $|\dot{\theta}_1|, |\dot{\theta}_2| \leq 2$ rad/s and the feasible joint accelerations are $|\ddot{\theta}_1|, |\ddot{\theta}_2| \leq 0.5$ rad/s². Find the fastest motion time T using a

cubic time scaling that satisfies the joint velocity and acceleration limits.

Exercise 9.5 Find the fifth-order polynomial time scaling that satisfies $s(T) = 1$ and $s(0) = \dot{s}(0) = \ddot{s}(0) = \dot{s}(T) = \ddot{s}(T) = 0$.

Exercise 9.6 As a function of the total time of motion T , find the times at which the acceleration $\ddot{s}$ of the fifth-order polynomial point-to-point time scaling is a maximum or a minimum.

Exercise 9.7 If you want to use a polynomial time scaling for point-to-point motion with zero initial and final velocities, accelerations, and jerks, what would be the minimum order of the polynomial?

Exercise 9.8 Prove that the trapezoidal time scaling, using the maximum allowable acceleration a and velocity v , minimizes the time of motion T .

Exercise 9.9 Plot by hand the acceleration profile $\ddot{s}(t)$ for a trapezoidal time scaling.

Exercise 9.10 If v and a are specified for a trapezoidal time scaling of a robot, prove that $v^2/a \leq 1$ is a necessary condition for the robot to reach the maximum velocity v during the path.

Exercise 9.11 If v and T are specified for a trapezoidal time scaling, prove that $vT > 1$ is a necessary condition for the motion to be able to complete in time T . Prove that $vT \leq 2$ is a necessary condition for a three-stage trapezoidal motion.

Exercise 9.12 If a and T are specified for a trapezoidal time scaling, prove that $aT^2 \geq 4$ is a necessary condition to ensure that the motion completes in time.

Exercise 9.13 Consider the case where the maximum velocity v is never reached in a trapezoidal time scaling. The motion becomes a bang-bang motion: constant acceleration a for time $T/2$ followed by constant deceleration $-a$ for time $T/2$. Write down the position $s(t)$, velocity $\dot{s}(t)$, and acceleration $\ddot{s}(t)$ for

both phases, in analogy to Equations (9.16)–(9.24).

Exercise 9.14 Plot by hand the acceleration profile $\ddot{s}(t)$ for an S-curve time scaling.

Exercise 9.15 A seven-stage S-curve is fully specified by the time t_J (the duration of a constant positive or negative jerk), the time t_a (the duration of constant positive or negative acceleration), the time t_v (the duration of constant velocity), the total time T , the jerk J , the acceleration a , and the velocity v . Of these seven quantities, how many can be specified independently?

Exercise 9.16 A nominal S-curve has seven stages, but it can have fewer if certain inequality constraints are not satisfied. Indicate which cases are possible with fewer than seven stages. Sketch by hand the $\dot{s}(t)$ velocity profiles for these cases.

Exercise 9.17 If the S-curve achieves all seven stages and uses a jerk J , an acceleration a , and a velocity v , what is the constant-velocity coasting time t_v in terms of v , a , J , and the total motion time T ?

Exercise 9.18 Write your own via-point cubic-polynomial interpolation trajectory generator program for a two-dof robot. A new position and velocity specification is required for each joint at 1000 Hz. The user specifies a sequence of via-point positions, velocities, and times, and the program generates an array consisting of the joint angles and velocities at every millisecond from time $t = 0$ to time $t = T$, the total duration of the movement. For a test case with at least three via points (one at the start and one at the end, both with zero velocity, and at least one more via point), plot

- (a) the path in the joint angle space (similar to Figure 9.7), and
- (b) the position and velocity of each joint as a function of time (these plots should look similar to Figure 9.8).

Exercise 9.19 Via points with specified positions, velocities, and accelerations can be interpolated using fifth-order polynomials of time. For a fifth-order polynomial segment between via points j and $j + 1$, of duration ΔT_j , with β_j , β_{j+1} , $\dot{\beta}_j$, $\dot{\beta}_{j+1}$, $\ddot{\beta}_j$, and $\ddot{\beta}_{j+1}$ specified, solve for the coefficients of the fifth-order polynomial (which is similar to Equations (9.26)–(9.29)). A symbolic math

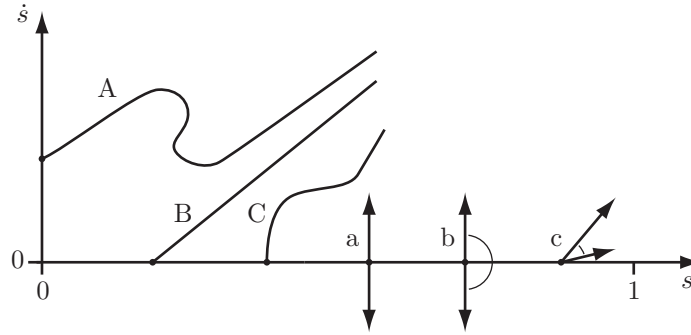


Figure 9.16: A, B, and C are candidate integral curves, originating from the dots indicated, while a, b, and c are candidate motion cones at $\dot{s} = 0$. Two of the integral curves and two of the motion cones are incorrect.

solver will simplify the problem.

Exercise 9.20 By hand or by computer, plot a trapezoidal motion profile in the $(s, \dot{s})$ -plane.

Exercise 9.21 Figure 9.16 shows three candidate motion curves in the $(s, \dot{s})$ -plane (A, B, and C) and three candidate motion cones at $\dot{s} = 0$ (a, b, and c). Two of the three curves and two of the three motion cones cannot be correct for any robot dynamics. Indicate which are incorrect and explain your reasoning. Explain why the remaining curve and motion cone are possibilities.

Exercise 9.22 Under the assumptions of Section 9.4.4, explain why the time-scaling algorithm of Section 9.4.2 (see Figure 9.13) is correct. In particular,

- explain why, in the binary search of step 4, the curve integrated forward from $(s_{\text{lim}}, \dot{s}_{\text{test}})$ must either hit (or run tangent to) the velocity limit curve or hit the $\dot{s} = 0$ axis (and does not hit the curve F , for example);
- explain why the final time scaling can only touch the velocity limit curve tangentially; and
- explain why the acceleration switches from minimum to maximum at points where the time scaling touches the velocity limit curve.

Exercise 9.23 Explain how the time-scaling algorithm should be modified to handle the case where the initial and final velocities, at $s = 0$ and $s = 1$, are

nonzero.

Exercise 9.24 Explain how the time-scaling algorithm should be modified if the robot's actuators are too weak to hold it statically at some configurations of the path (the static-posture-maintenance assumption is violated), but the assumptions on inadmissible states and zero-inertia points are satisfied. Valid time scalings may no longer exist. Under what condition(s) should the algorithm terminate and indicate that no valid time scaling exists? (Under the assumptions of Section 9.4.4 the original algorithm always finds a solution and therefore does not check for failure cases.) What do the motion cones look like at states $(s, \dot{s} = 0)$ where the robot cannot hold itself statically?

Exercise 9.25 Create a computer program that plots the motion cones in the $(s, \dot{s})$ -plane for a 2R robot in a horizontal plane. The path is a straight line in joint space from $(\theta_1, \theta_2) = (0, 0)$ to $(\pi/2, \pi/2)$. Use the dynamics from Equation (8.9) (with $g = 0$), then rewrite the dynamics in terms of $s, \dot{s}, \ddot{s}$ instead of $\theta, \dot{\theta}, \ddot{\theta}$. The actuators can provide torques in the range $-\tau_{i,\text{limit}} - b\dot{\theta}_i \leq \tau_i \leq \tau_{i,\text{limit}} - b\dot{\theta}_i$, where $b > 0$ indicates the velocity dependence of the torque. The cones should be drawn at a grid of points in $(s, \dot{s})$. To keep the figure manageable, normalize each cone ray to the same length.

Exercise 9.26 We have been assuming forward motion on a path, $\dot{s} > 0$. What if we allowed backward motion on a path, $\dot{s} < 0$? This exercise involves drawing motion cones and an integral curve in the $(s, \dot{s})$ -plane, including both positive and negative values of $\dot{s}$. Assume that the maximum acceleration is $U(s, \dot{s}) = U > 0$ (constant over the $(s, \dot{s})$ -plane) and the maximum deceleration is $L(s, \dot{s}) = L = -U$. You can assume, for example, that $U = 1$ and $L = -1$.

- For any constant s , draw the motion cones at the five points where $\dot{s}$ takes the values $\{-2, -1, 0, 1, 2\}$.
- Assume the motion starts at $(s, \dot{s}) = (0, 0)$ and follows the maximum acceleration U for time t . Then it follows the maximum deceleration L for time $2t$. Then it follows U for time t . Sketch by hand the integral curve. (The exact shape does not matter, but the curve should have the correct features.)